

# Internship Notification Form, IIT Delhi

## About Organisation

|                                 |                                                           |
|---------------------------------|-----------------------------------------------------------|
| <b>Name of Company:</b>         | Micron Technology                                         |
| <b>Date of Establishment:</b>   | 1978-10-05                                                |
| <b>Number of Employees:</b>     | 48000                                                     |
| <b>Social Media Page Link:</b>  | <a href="http://www.micron.com">http://www.micron.com</a> |
| <b>Website:</b>                 | <a href="http://www.micron.com">http://www.micron.com</a> |
| <b>Type of Organization:</b>    | MNC (Foreign Origin)                                      |
| <b>Location of Head office:</b> | Boise, Idaho                                              |
| <b>Nature of Business:</b>      | Other (core_engineering,manufacturing)                    |

## Internship Profile

**Job Title:** Intern,Central Engineering-CXL Engineer

**Job Description:**

About Micron: We are an industry leader in innovative memory and storage solutions transforming how the world uses information to enrich life for all. With a relentless focus on our customers, technology leadership, and manufacturing and operational excellence, Micron delivers a rich portfolio of high-performance DRAM, NAND and NOR memory and storage products through our Micron® and Crucial® brands. At Micron India, you will be a part of one of the fastest ramps in Indian semiconductor history. We are ranked among Top 50 organizations for India's Best Workplaces in Manufacturing 2024 rankings by Great Place to Work Institute. In about five years, we have built an operations footprint with 20 centres of excellence and grown to over 3500 team members. We nurture a culture that is fair, fearless and fun that works to enhance personal and professional development at every level.

Culture and Values at Micron: As a leader in memory solutions our team members, partners, customers, and leadership work closely together to deliver — setting a high bar for ourselves and our industry. We believe transformation always starts within and our core values are driven by the following 5 pillars, namely:

- People - We care about each other.
- Innovation - We develop solutions that shape the world's future.
- Tenacity - Nothing shakes our resolve.
- Collaboration - We work as one team.
- Customer Focus - We win by knowing our customers

About CXL Engineering team:

The Compute Express Link (CXL) team is committed to building Micron's CXL products. The team's work includes enhancing the Linux Kernel for CXL, developing and optimizing system software that manages and utilizes CXL devices, designing and implementing test solutions using Python to automate testing processes, validate system performance, and ensure the reliability of CXL products among many things

#### Profile 1

About profile – Intern - CXL Kernel Engineer: As a system software engineer in the system software team, you will have the opportunity to work on critical projects that enhance the performance and reliability of Micron's CXL products. This role involves hands-on experience with Linux kernel development, system software optimization, and the creation of automated test solutions

Responsibilities will include, but are not limited to:

- Assist in enhancing Linux kernel support for CXL devices, ensuring seamless integration and optimal performance.
- Contribute to the development and optimization of system software that manages and utilizes CXL devices, focusing on memory management and bus interface protocols.
- Design and implement automated test solutions using Python to validate system performance and ensure the reliability of CXL products
- Security: Work on enhancing memory security, addressing vulnerabilities, and implementing security measures to protect against memory-based attacks.
- Documentation: Maintain comprehensive documentation of memory management components, algorithms, and processes.

#### Qualifications:

- Strong understanding of Linux operating systems and kernel internals, particularly memory management and PCIe driver ecosystem.
- Proficiency in Rust and C/C++.
- Strong understanding of scripting language like Python, TCL or PERL.
- Familiarity with full stack development.
- Familiarity with hardware memory management units (MMUs) and cache architectures.
- Familiarity with memory analysis tools and profilers.
- Experience with debugging tools and techniques, as well as performance optimization.
- Knowledge of memory management algorithms, virtual memory, and page replacement policies.
- Familiarity with operating system concepts, system calls, and synchronization mechanisms.
- Understanding of hardware architecture and its interaction with memory.
- Excellent problem-solving skills and the ability to work collaboratively in a team.
- Strong communication skills both written and verbal.
- Strong interpersonal skills and maintain positive relationships.
- Ambitious and goal oriented.

Location:  
Bangalore

Education Requirement & eligibility:

- B.Tech degree in Computer Science or a related field is preferred.
- CGPA requirements = 7.0 CGPA & Above

Profile 2

About profile – Intern - CXL Software Development Engineer: As an Intern CXL Software Engineer at Micron Technology Inc., you'll be part of a team working on cutting-edge memory interface technology known as Compute Express Link (CXL). This technology is revolutionizing the world of computing. Our goal is to provide interns with hands-on engineering projects, allowing them to gain valuable experience in CXL technology, memory systems, and the product life cycle while making a meaningful impact on Micron's business.

Our intern programs are designed to prepare engineers for future roles and offer an extraordinary experience that complements their academic studies. You'll collaborate with various teams, including firmware, validation and system enablers both within and outside of Micron. Your day-to-day tasks may involve software development for Micron's CXL modules, developing innovative test methodologies, data analysis, creating custom tools, and effectively communicating findings to stakeholders.

While you'll be based in either the Micron office in Bangalore, India, you'll work closely with experienced Micron Engineers from both India and the United States. Flexibility in working hours will be essential. The Software Engineer role is critical to Micron's mission of producing the most innovative and differentiated CXL products. This internship opportunity is available for either a 2- or 6-month duration, with the potential for a future career path at Micron in India.

Responsibilities will include, but are not limited to:

- Proficiency in C/C++/Java and/or Python.
- Define and Automate data collection process for given problem statements.
- Ability to drive innovation and best in class solutions for Micron products.
- Evaluate power and performance based on Benchmark and In-house workloads.
- Collaborate with firmware validation teams to bring in issues and root cause them.

Preferred Skills:

- Demonstrating the ability to tackle complex problems efficiently.
- Desirable knowledge in Python, C, and C++ programming or Java, OOPS concepts as well as understanding embedded systems, computer architecture, and memory systems.
- Good capability to solve problems on Data structures and algorithms.
- Having a good grasp of machine learning and artificial intelligence concepts.
- Knowledge of DBMS internals and concepts on Data Warehousing.
- Preferably having contributed to Linux open source projects.
- Creating personal projects independently on GitHub.
- Familiarity with version control software (e.g., Git).
- Knowing the Linux operating system and basic shell scripting.

- scripting.
- Excellent communication skills.
- Being proactive and passionate about learning.
- Maintaining positive relationships with colleagues.
- Having clear goals and a drive to achieve them.
- Working effectively in dynamic team environments.

Location:  
Bangalore

Education Requirement & eligibility:

- B.Tech/M.Tech degree in Computer Science/ Computer Engineering or a related field is preferred.
- CGPA requirements = 7.0 CGPA & Above

Micron by Numbers:

- \$15.5B FY23 annual revenue
- 136 on the 2023 Fortune 500
- 55,000+ patents granted and growing
- Operations in 17 countries
- 11 manufacturing sites and 12 customer labs
- 20 design center locations
- ~44,000 team members
- Official Website: <http://www.micron.com>

|                                             |           |
|---------------------------------------------|-----------|
| <b>Minimum No. of Hires:</b>                | 1         |
| <b>Expected No. of Hires:</b>               | 1         |
| <b>Location(s)/Place of Posting/Online:</b> | Bangalore |
| <b>Skillset:</b>                            | Refer JD  |
| <b>Minimum CGPA:</b>                        | 7         |
| <b>Students with backlog eligible:</b>      | No        |

## Selection Process

|                                          |         |
|------------------------------------------|---------|
| <b>Resume Shortlist:</b>                 | Yes     |
| <b>Mode of Selection:</b>                | Virtual |
| <b>Resume shortlisting before test?:</b> | Yes     |
| <b>Test:</b>                             | Yes     |
| <b>Mode of Test:</b>                     | Offline |
| <b>Test duration (minutes):</b>          | 90      |
| <b>Aptitude/Psycometric:</b>             | Yes     |
| <b>Technical:</b>                        | Yes     |
| <b>Group Discussion:</b>                 | No      |
| <b>Personal Interview:</b>               | No      |
| <b>Technical Round:</b>                  | Yes     |

|                                  |    |
|----------------------------------|----|
| <b>HR Round:</b>                 | No |
| <b>Other Processes (if any):</b> |    |
| <b>No. of Rounds:</b>            | 1  |
| <b>Medical Test:</b>             | No |

## Eligible Academic Programs

|                              |                                                                                                                                                                                                                                                                           |
|------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Diversity Recruiting:</b> | No                                                                                                                                                                                                                                                                        |
| <b>Eligible Years:</b>       | Graduating in 2026 (Pre-Final Year Students) - B.Tech / Dual / Master's                                                                                                                                                                                                   |
| <b>Eligible Departments:</b> | B.Tech in Computer Science & Engineering, B.Tech in Electrical Engineering, B.Tech in Electrical Engineering (Power and Automation), B.Tech in Mathematics & Computing, B.Tech and M.Tech in Computer Science & Engineering, B.Tech and M.Tech in Mathematics & Computing |

## Stipend Details

|                                                |                                                                                               |
|------------------------------------------------|-----------------------------------------------------------------------------------------------|
| <b>Stipend (per month) (In INR Per Month):</b> | 50,000 INR Per Month                                                                          |
| <b>Any other perks/ benefits/ components:</b>  | Relocation assistance provided subject to the candidate meeting the eligibility requirements. |